

Data Mining Course Project

1. Objective

The objective of this project is to enable students to apply data mining techniques on a real-world dataset, compare multiple methods, evaluate their performance using appropriate metrics, and present their work in the form of a scientific paper.

2. Project Topic

Each student or group must select **one** of the following data mining tasks:

- Classification
- Clustering
- Regression
- Association Rule Mining
- Anomaly Detection
- Dimensionality Reduction
- Time Series Analysis

The selected task must be appropriate for the chosen dataset.

3. Dataset Selection

Students must select their dataset from reliable public sources such as:

- UCI Machine Learning Repository
- Kaggle
- or similar platforms

Requirements:

- The dataset should be **relatively recent** (not outdated).
- It must be suitable for the selected data mining task.
- It should be large and meaningful enough to support proper analysis.
- Very small or trivial datasets may not be accepted.
- The dataset source must be clearly specified.

4. Methodology Requirements

Students are required to apply **multiple data mining methods** on the same dataset.

Minimum expectations:

- At least **two different methods** must be implemented

- Methods must be compared on the same dataset
- Data preprocessing steps must be clearly explained
- The rationale behind method selection must be justified

5. Evaluation Requirements

Projects must be evaluated using **multiple evaluation metrics**, depending on the selected task.

Examples:

For Classification:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

For Clustering:

- Silhouette Score
- Davies-Bouldin Index
- Calinski-Harabasz Index

For Regression:

- MAE
- MSE
- RMSE
- R^2

Students must justify their choice of evaluation metrics.

6. Paper Format

The project must be written in the form of an **academic paper**.

The paper must include the following sections:

- Title
- Abstract

- Keywords
- Introduction
- Related Work
- Dataset and Preprocessing
- Proposed Method / Methodology
- Experimental Results and Discussion
- Conclusion
- References

7. Expected Content

The paper should clearly include:

- Problem definition
- Dataset description
- Preprocessing steps
- Description of applied methods
- Parameter settings
- Experimental setup
- Evaluation metrics
- Results (tables/figures)
- Comparative analysis
- Discussion of findings

8. Mandatory Requirements

- A valid data mining task must be selected
- A suitable dataset must be used
- Multiple methods must be applied
- Multiple evaluation metrics must be used
- The project must be written in paper format
- All sources must be properly cited
- Tools and libraries must be specified

9. Bonus (Extra Credit – 30%)

The following are **not mandatory**, but will contribute to the bonus score (**30% of the total grade**):

- Writing the paper using **LaTeX**
- **Turnitin similarity score below 20%**
- Proposing a **novel/improved method or idea**
- Comparing results with **existing literature**
- Preparing the paper at a **conference/journal submission level**
- Providing deeper analysis and stronger experimental design

10. Academic Integrity

- All work must be original
- Proper citation is required
- Plagiarism will result in rejection
- Students are responsible for all submitted content

11. Submission Requirements

Students must submit:

- Final paper (PDF)
- Source code
- Dataset link
- (Optional) LaTeX source files

12. Presentation

- Presentations will be held on **May 23–24** in **two sessions**
- Each session will start at **13:00**
- Presentations will be conducted according to the **attendance list order**

13. Group Work

- Projects can be done **individually or in groups of up to 2 students**
- Each student is expected to contribute equally

14. Evaluation Criteria

Mandatory Part – 70%

- Dataset and problem selection: 10%

- Preprocessing and methodology: 15%
- Application of multiple methods: 15%
- Evaluation metrics usage: 10%
- Results and discussion: 10%
- Paper format and writing quality: 10%

Bonus Part – 30%

- LaTeX usage: 5%
- Turnitin (<20%): 5%
- Novel contribution: 10%
- Publication-quality writing: 10%

15. Notes

- Simple or superficial projects will receive low grades
- Running algorithms is not sufficient; interpretation is required
- Visual quality, clarity, and academic writing matter
- Method-task compatibility is important